

5.4.3 Cholesky factorization algorithm (right-looking variant)

05.4.3 Cholesky factorization algorithm



The most common algorithm for computing the Cholesky factorization of a given HPD matrix A is derived as follows:

- Consider $A = LL^H$, where L is lower triangular.

Partition

$$A = \begin{pmatrix} \alpha_{11} & \star \\ a_{21} & A_{22} \end{pmatrix} \quad \text{and} \quad L = \begin{pmatrix} \lambda_{11} & 0 \\ l_{21} & L_{22} \end{pmatrix}. \quad (5.4.1)$$

Since A is HPD, we know that

- α_{11} is a positive number ([Homework 5.4.1.2](#)).
- A_{22} is HPD ([Homework 5.4.1.3](#)).
- By substituting these partitioned matrices into $A = LL^H$ we find that

$$\begin{pmatrix} \alpha_{11} & \star \\ a_{21} & A_{22} \end{pmatrix} = \begin{pmatrix} \lambda_{11} & 0 \\ l_{21} & L_{22} \end{pmatrix} \begin{pmatrix} \lambda_{11} & 0 \\ l_{21} & L_{22} \end{pmatrix}^H = \begin{pmatrix} \lambda_{11} & 0 \\ l_{21} & L_{22} \end{pmatrix} \begin{pmatrix} \bar{\lambda}_{11} & l_{21}^H \\ 0 & L_{22}^H \end{pmatrix} \\ = \begin{pmatrix} |\lambda_{11}|^2 & \star \\ \lambda_{11}l_{21} & l_{21}l_{21}^H + L_{22}L_{22}^H \end{pmatrix},$$

from which we conclude that

$$\alpha_{11} = |\lambda_{11}|^2 \quad \star \\ a_{21} = \lambda_{11}l_{21} \quad A_{22} = l_{21}l_{21}^H + L_{22}L_{22}^H$$

or, equivalently,

$$\lambda_{11} = \pm\sqrt{\alpha_{11}} \quad \star \\ l_{21} = a_{21}/\bar{\lambda}_{11} \quad L_{22} = \text{Chol}(A_{22} - l_{21}l_{21}^H)$$

- These equalities motivate the following algorithm for overwriting the lower triangular part of A with the Cholesky factor of A :
 - Partition $A \rightarrow \begin{pmatrix} \alpha_{11} & \star \\ a_{21} & A_{22} \end{pmatrix}$.
 - Overwrite $\alpha_{11} := \lambda_{11} = \sqrt{\alpha_{11}}$. (Picking $\lambda_{11} = \sqrt{\alpha_{11}}$ makes it positive and real, and ensures uniqueness.)
 - Overwrite $a_{21} := l_{21} = a_{21}/\lambda_{11}$.
 - Overwrite $A_{22} := A_{22} - l_{21}l_{21}^H$ (updating only the lower triangular part of A_{22}). This operation is called a *symmetric rank-1 update*.
 - Continue by computing the Cholesky factor of A_{22} .

The resulting algorithm is often called the "right-looking" variant and is summarized in [Figure 5.4.3.1](#).

$A = \text{Chol-right-looking}(A)$
$A \rightarrow \left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right)$
A_{TL} is 0×0
while $n(A_{TL}) < n(A)$
$\left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \rightarrow \left(\begin{array}{c c} A_{00} & a_{01} \quad A_{02} \\ \hline a_{10}^T & \alpha_{11} \quad a_{12}^T \\ A_{20} & a_{21} \quad A_{22} \end{array} \right)$
$\alpha_{11} := \lambda_{11} = \sqrt{\alpha_{11}}$
$a_{21} := l_{21} = a_{21}/\alpha_{11}$
$A_{22} := A_{22} - a_{21}a_{21}^H$ (syr: update only lower triangular part)
$\left(\begin{array}{c c} A_{TL} & A_{TR} \\ \hline A_{BL} & A_{BR} \end{array} \right) \leftarrow \left(\begin{array}{c c} A_{00} & a_{01} \quad A_{02} \\ \hline a_{10}^T & \alpha_{11} \quad a_{12}^T \\ A_{20} & a_{21} \quad A_{22} \end{array} \right)$
endwhile

Figure 5.4.3.1. Cholesky factorization algorithm (right-looking variant). The operation "syr" refers to "symmetric rank-1 update", which performs a rank-1 update, updating only the lower triangular part of the matrix in this algorithm.

Homework 5.4.3.1. Give the approximate cost incurred by the algorithm in [Figure 5.4.3.1](#) when applied to an $n \times n$ matrix.

▼ [Answer](#) ▼ [Solution](#)

05.4.3 Cost of Cholesky factorization



The cost of the Cholesky factorization of $A \in \mathbb{C}^{n \times n}$ can be analyzed as follows: In [Figure 5.4.3.1](#) during the k th iteration (starting k at zero) A_{00} is $k \times k$. Thus, the operations in that iteration cost

- $\alpha_{11} := \sqrt{\alpha_{11}}$: this cost is negligible when k is large.
- $a_{21} := a_{21}/\alpha_{11}$: approximately $(n - k - 1)$ flops. This operation is typically implemented as $(1/\alpha_{11})a_{21}$.
- $A_{22} := A_{22} - a_{21}a_{21}^H$ (updating only the lower triangular part of A_{22}): approximately $(n - k - 1)^2$ flops.

Thus, the total cost in flops is given by

$$\begin{aligned}
 C_{\text{Chol}}(n) &\approx \text{< sum over all iterations >} \\
 &\quad \underbrace{\sum_{k=0}^{n-1} (n - k - 1)^2}_{\text{(Due to update of } A_{22})} + \underbrace{\sum_{k=0}^{n-1} (n - k - 1)}_{\text{(Due to update of } a_{21})} \\
 &= \text{< change of variables } j = n - k - 1 > \\
 &\quad \sum_{j=0}^{n-1} j^2 + \sum_{j=0}^{n-1} j \\
 &\approx \text{< } \sum_{j=0}^{n-1} j^2 \approx n^3/3; \sum_{j=0}^{n-1} j \approx n^2/2 > \\
 &\quad \frac{1}{3}n^3 + \frac{1}{2}n^2 \\
 &\approx \text{< remove lower order term >} \\
 &\quad \frac{1}{3}n^3.
 \end{aligned}$$

$\frac{1}{3}n^3$ flops.

Remark 5.4.3.2. Comparing the cost of the Cholesky factorization to that of the LU factorization in [Homework 5.2.2.1](#), we see that taking advantage of symmetry cuts the cost approximately in half.

Homework 5.4.3.2. Implement the algorithm given in [Figure 5.4.3.1](#) as

```
function [ A_out ] = Chol_right_looking( A )
```

by completing the code in [Assignments/Week05/matlab/Chol_right_looking.m](#).

Input is a HPD $m \times n$ matrix A with only the lower triangular part stored. Output is the matrix A that has its lower triangular part overwritten with the Cholesky factor. You may want to use

[Assignments/Week05/matlab/test_Chol_right_looking.m](#) to check your implementation.

▼ [Solution](#)

See [Assignments/Week05/answers/Chol_right_looking.m](#).
(Assignments/Week05/answers/Chol_right_looking.m)